BrainBolt - Answer Key

Topic: Python (15 Questions)

Q1. Which keyword is used to define a function in Python?

A. func

B. def <- correct

C. function

D. lambda

Answer: B

Why: 'def' is the reserved keyword used to declare a function in Python.

Q2. What is the output of: print(2 ** 3 ** 2) ?

A. 64

B. 512 <- correct

C. 256

D. 12

Answer: B

Why: The ** operator is right-associative, so it evaluates as 2 ** (3 ** 2) = 2 ** 9 = 512.

Q3. Which data type is mutable in Python?

A. tuple

B. string

C. list <- correct

D. frozenset

Answer: C

Why: Lists are mutable — you can change, add, or remove items after creation.

Q4. What does the 'self' keyword represent inside a class method?

A. The class itself

B. The current instance of the class <- correct

C. The parent class

D. A static variable

Answer: B

Why: 'self' refers to the current instance of the class on which the method is being called.

Q5. Which of the following is NOT a valid Python data type?

A. dict

B. set

C. array <- correct

D. tuple

Answer: C

Why: 'array' is not a built-in primitive type — it must be imported from the 'array' module. dict, set and tuple are built-in.

Q6. What will len('Python') return?

A. 5

B. 6 <- correct

C. 7

D. Error

Answer: B

Why: The string 'Python' has 6 characters: P, y, t, h, o, n.

Q7. Which method is used to add an element at the end of a list?

A. add()

B. insert()

C. append() <- correct

D. push()

Answer: C

Why: list.append(x) adds element x to the end of the list.

Q8. What is the correct way to handle exceptions in Python?

A. try-except <- correct

B. try-catch

C. do-catch

D. handle-error

Answer: A

Why: Python uses the try-except construct for exception handling, not try-catch.

Q9. Which symbol is used for single-line comments in Python?

A. //

B. # <- correct

C. /* */

D. --

Answer: B

Why: The # symbol starts a single-line comment in Python.

Q10. What does the 'pass' statement do?

A. Exits a loop

B. Skips current iteration

C. Does nothing — placeholder <- correct

D. Raises an exception

Answer: C

Why: 'pass' is a null statement used as a placeholder where code is syntactically required but no action is desired.

Q11. Which built-in function converts a string to an integer?

A. str()

B. int() <- correct

C. float()

D. input()

Answer: B

Why: int('42') converts the string '42' to the integer 42.

Q12. What is the output of: print(type([])) ?

A. <class 'list'> <- correct

B. <class 'tuple'>

C. <class 'dict'>

D. <class 'set'>

Answer: A

Why: [] denotes an empty list, so its type is <class 'list'>.

Q13. Which OOP concept allows a child class to redefine a method of its parent class?

A. Encapsulation

B. Abstraction

C. Method Overriding <- correct

D. Polymorphism via overloading

Answer: C

Why: Method overriding lets a subclass provide a specific implementation of a method already defined in its superclass.

Q14. What does the 'in' operator do?

A. Imports a module

B. Checks membership in a sequence <- correct

C. Defines an iterator

D. Initialises a variable

Answer: B

Why: The 'in' operator returns True if a value is found in the given sequence (list, string, tuple, etc.).

Q15. Which keyword is used to create a generator function?

A. return

B. yield <- correct

C. gen

D. produce

Answer: B

Why: Functions that use 'yield' instead of 'return' become generators — they produce values lazily.